

Titanic Passenger Analysis Report

Executive Summary

This report accompanies the Titanic Interactive Dashboard, providing an overview of the passenger demographics and survival rates aboard the RMS Titanic. The interactive dashboard allows for real-time filtering and visualization of the dataset, making it easy to uncover patterns related to passenger class, gender, age, and embarkation points.

Dataset Overview

The dataset contains information on 1,309 passengers, including their socio-economic status (Ticket Class), age, gender, fare paid, and whether they survived the tragic sinking.

Key attributes analyzed include:

- Passenger Class (Pclass): 1st, 2nd, and 3rd class.
- Gender (Sex): Male and Female.
- Age: Passenger's age in years.
- Embarked: Port of embarkation (Southampton, Cherbourg, Queenstown).
- Survived: Survival status (1 = Survived, 0 = Perished).

Key Insights from the Dashboard

1. Overall Survival Rate

The overall survival rate across the dataset is roughly 38.2%, meaning a significant majority of the passengers did not survive. The dashboard prominently displays this KPI, updating dynamically based on the applied filters.

2. Gender and Survival (Women and Children First)

The "Women and children first" protocol had a massive impact on survival rates. Females had a dramatically higher survival rate compared to males. The Gender Distribution doughnut chart visualizes the stark contrast, showing that the majority of survivors were female, while the vast majority of casualties were male.

3. Socio-Economic Status (Passenger Class)

Passenger class strongly correlated with survival. 1st Class passengers had the highest survival rate, significantly outperforming 2nd and 3rd classes. 3rd Class passengers had the highest number of casualties. The Survival by Passenger Class bar chart perfectly illustrates this socio-economic disparity.

4. Age Distribution

The highest concentration of passengers was in the 21-30 age group. Younger age brackets (especially children aged 0-10) had relatively high survival rates compared to middle-aged males in lower ticket classes. The Survival by Age Group chart provides a stacked view of how different age demographics fared.

5. Embarkation Ports

Most passengers boarded the Titanic at Southampton (S). Passengers embarking from Cherbourg (C) had a proportionately higher survival rate, which correlates with a larger percentage of 1st-class passengers boarding at that port.

Using the Dashboard

The accompanying `Titanic_Interactive_Dashboard.html` is a standalone, single-file application.

Features:

- **Dynamic KPI Cards:** Automatically calculate total passengers, average age, average fare, and survival rate based on current filters.
- **Interactive Charts:** Hover over the charts to view exact data points and tooltips.
- **Live Filtering:** Use the dropdowns at the top to filter the entire dashboard by Passenger Class, Gender, or Embarkation Port.
- **Data Table:** A snapshot table at the bottom showing a sample of the raw passenger records currently in view.
- **How to Share:** Because all data and styling are bundled into this single HTML file, you can easily share the dashboard by sending the `Titanic_Interactive_Dashboard.html` file via email or any messaging app. Anyone can open it using a standard web browser (Chrome, Edge, Safari) without needing to install anything or connect to a database.